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BRIEF REPORT

Physiology of Maternal Separation in a Bonnet Macaque Infant

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A bonnet monkey infant (*M. radiata*) was monitored both physiologically and behaviorally during a separation experiment in which both its mother and other conspecifiecs were removed from its social group leaving the bonnet infant with three adult pigtailed females. The separated infant exhibited the characteristic slouched posture of depression and physiological changes including decreases in the heart rate and body temperature, increases in cardiac arrhythmias, and alterations in sleep patterns, as has been previously described in separated pigtailed infants. Persistent decreases in heart rate and body temperature were noted following reunion with the mother, possibly reflecting a disturbance in the mother-infant relationship secondary to the mother's having come into estrus during the period of separation. This study indicates that the physiological correlates of maternal separation previously described in pigtailed infants are not species specific.

Key words: separation, monkeys, depression, *M. radiata*, mother-infant, species specificity, physiology, biotelemetry, attachment, heart rate, body temperature, sleep patterns

INTRODUCTION

Previous studies have demonstrated that social group living pigtailed (*M. nemestrina*) monkey infants who are separated from their mothers experience an "agitation-depression" or "protest-despair" behavioral reaction [Kaufman and Rosenblum, 1967] as well as characteristic physiological changes during the period of behavioral depression, including disturbances in heart rate and rhythm, decreases in nocturnal body temperature, and pronounced disturbances in sleep patterns with increases in time awake and number of arousals, and decreases in REM sleep [Reite et al, 1981]. We and others [Bowlby, 1969; Harlow et al, 1972; Reite, 1977] have suggested that this organismic reaction to the disruption of an attachment bond is likely homologous with the protest-despair, or anaclitic depression seen in human infants who have been separated from their mothers, or who have lost proper mothering, and that the physio-

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logical data obtained from nonhuman primates may contribute directly to our understanding of the pathophysiology of separation and loss in humans [Reite et al, 1981]. Since quantified physiological observations have been available only for the pigtailed macaque, it has not been possible to determine to what extent this constellation of physiological changes accompanying maternal separation in monkeys might be species specific.

Bonnet macaques (*M. radiata*) exhibit very different social behavioral patterns than do pigtailed macaques, with bonnets exhibiting more social cohesiveness and more prominent "aunting" behavior in infant rearing [Kaufman and Rosenblum, 1969a]. Previous studies of maternal separations, in which the separated bonnet infant was left in its social group, resulted in the separated infant being "adopted" by other adults, resulting in marked attenuation of the behavioral reaction to separation [Kaufman and Rosenblum, 1969b]; however, social group-living bonnet infants may become depressed following a maternal separation in a high foraging demand environment [Plimpton and Rosenblum, 1980]. Kaufman and Stynes [1978] reported a seminal experiment in which a bonnet infant, raised in a group containing both bonnets and pigtailed monkeys, was separated from its mother and other conspecifics and left in the group with the familiar pigtailed monkeys. This infant exhibited a behavioral depressive reaction equally as severe as that seen in pigtailed infants after separation from the mother only. Physiological data were not recorded from that infant, however. We now present data obtained from another bonnet infant that demonstrates that the characteristic physiological reaction to maternal separation seen in pigtailed monkeys is not specific to that species, but may be observed in bonnet monkeys as well.

METHODS

During December, 1980 and January, 1981, we composed a social group consisting of 3 adult female pigtailed macaques without infants, 5 adult female bonnet macaques (3 with infants), and one adult bonnet male. The subject of this experiment, animal number B6.7.2, was born on 9-05-80 while in another bonnet social group, and was added, with his mother, to the mixed species group. At the age of 127 days, the subject and his mother were removed from the group, and, under general anesthesia, the subject was implanted with a hybrid multichannel biotelemetry system that transmitted eye movement (EOG), muscle activity (EMG), 3 channels of extradural EEG, body temperature (BT), and electrocardiogram (EKG) [Pauley and Reite, 1981]. After recovery from surgery, the subject and his mother were returned to the social group and 24-hr a day physiological recordings, as well as six 5-min quantified behavioral observations daily began [Reite and Short, 1980]. Behavioral variables included a measure [developed by Hinde and Atkinson, 1970] of "tension" in the mother-infant relationship. Based upon who (infant or mother) initiates the leaves and aproaches during the behavioral observations, the "tension" scores could vary from −1 (mother responsible for maintaining proximity or contact) to +1 (infant responsible for maintaining proximity or contact), with positive scores interpreted as measures of "tension" in the relationship (from the infant's standpoint). All behavioral observations were pooled by day. Physiological data were processed so as to obtain measures including mean daytime (1000-1600) and mean nighttime (2200-0400) heart rate (HR) and BT, and scores for several nocturnal sleep pattern variables each night.

Following the collection of 6 days of baseline (preseparation) behavioral and physiological data, the subject's mother, and all bonnet conspecifics, were removed from the social group, leaving the subject with the 3 adult female pigtails. After 10 complete days of separation, all bonnets (including the mother) were returned to the group, and behavioral and physiological recordings continued for another 5 days.

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RESULTS

Immediately after the separation, the bonnet exhibited behavioral agitation, increased vocalization, and active searching behavior. During this time HR and BT were markedly elevated, and the EEG indicated an active alert pattern. Behaviorally, the infant began exhibiting the characteristic slouched posture of depression on the second complete day of separation (see Fig. 1); this slouched behavior increased to 41% of total time observed (%TTO) by the eighth day of separation (see Fig. 2). Efforts on the part of the infant to contact the adult female pigtails were generally met with indifference, although one pigtail female did allow the infant to sleep next to her at night, and later in the separation period, to spend considerable time in physical contact with her during the day. Observers noted emergence of a "sad" facial expression and slowing of locomotion beginning the fourth full day of separation. The amount of time spent eating remained at or above baseline levels during separation, and distress vocalization occurred throughout separation.

Noctural HR varied about a mean of 182 bpm during baseline, and decreased precipitously during separation with nocturnal mean values ranging between 120 and 130 bpm late in the separation period, a decrease of about 50 bpm (Fig. 2). Night HR increased to near baseline levels the first night of reunion, and then decreased to separation levels later during reunion. The number of cardiac arrhythmias increased precipitously during separation. Nocturnal BT decreased by nearly 1 °C the first night of separation, returning to levels above baseline the first night of reunion, but then slowly decreasing again to separation levels. Mean daytime HR also decreased during separation and after 1 day of reunion. Mean day BT was not significantly changed during either separation or reunion. Nocturnal sleep changes during separation included increases in the time awake and decreases in REM time.

Following reunion, the infant's mother spent a great deal more time in reestablishing her social position, and in attracting the attention of the male, (she had gone into estrus during the period of separation) than in mothering her infant, although she did permit some physical contact and did allow the infant to sleep with her in ventral-ventral contact all night. The time the infant spent enclosed by the mother decreased from a mean of 25% TTO during baseline to 7.9% during reunion. The "tension" score increased from .38 ± .21 during baseline to .71 ± .19 during reunion. Nocturnal sleep patterns returned to baseline mean values after reunion. The first night after the mother returned, the infant's HR and BT returned to normal baseline values, but thereafter HR and BT mean values again declined to values similar to those seen during separation until the 5th day of reunion, when physiological recordings terminated. Follow-up physiological recordings were performed at intervals up to 12 weeks after reunion. The infant's BT had returned to baseline value by 13 days after reunion, but HR remained slightly lower than baseline at that time. Both HR and BT had returned to baseline values 5 weeks following reunion.

DISCUSSION

The physiological changes (in HR, BT, and sleep patterns) accompanying the behavioral depressive reaction in this separated and depressed bonnet infant are the same as those previously described as accompanying maternal separation and behavioral depression in pigtailed infants [Reite et al., 1981]. The manner in which the infant responded to reunion is most instructive. Heart rate and body temperature values, after a transient initial recovery the first night of reunion, returned to separation values, possibly reflecting the persisting alteration of the mother–infant relationship secondary to the mother's being in estrus, and reflected by the decreases in enclosed scores and increases in the "tension" value after reunion. Thus the infant was

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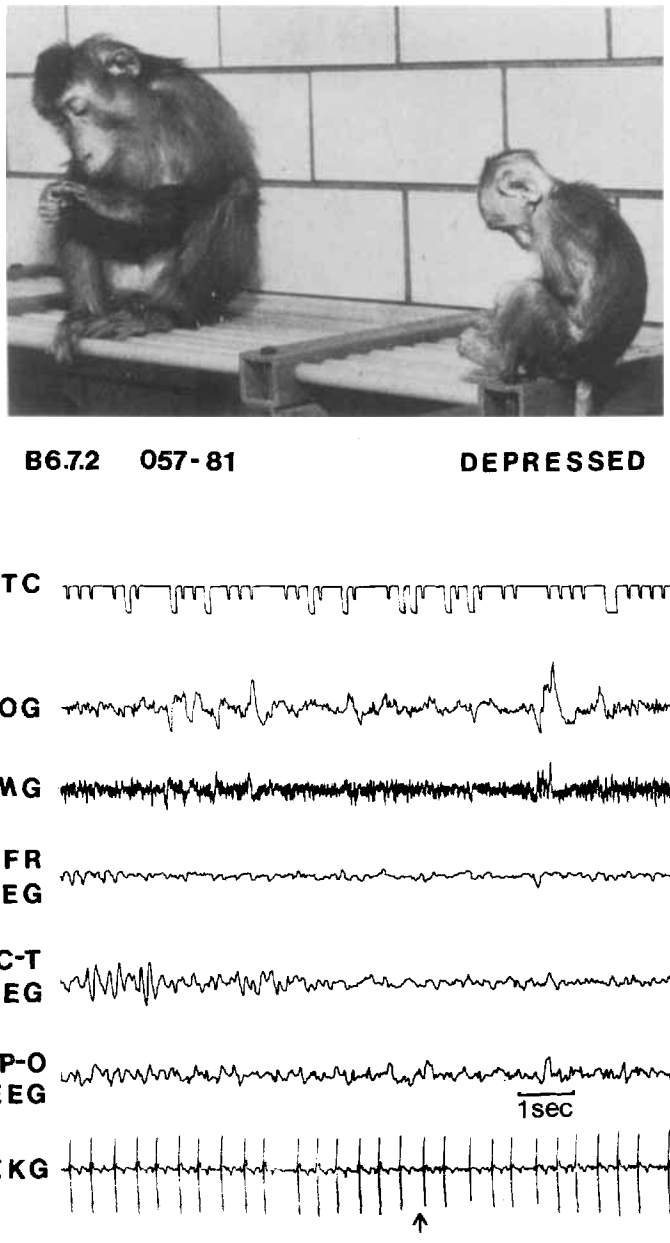


Fig. 1. Separated bonnet infant exhibiting the characteristic slouched posture of depression while an adult pigtailed female sits unconcerned nearby. TC = time code; EOG = eye movement; EMG = muscle activity; FR EEG = frontal EEG; C-T EEG = central temporal EEG; P-O EEG = parieto-occipital EEG; EKG = electrocardiogram. The picture was taken at the time indicted by the small arrow beneath the EKG trace.

in a sense reacting physiologically to the persisting disruption of the mother-infant attachment bond, even though the mother was physically present.

While little systematic data are available on the effects of maternal estrus per se on the physiological status of the infant, in our limited experience such effects seem to

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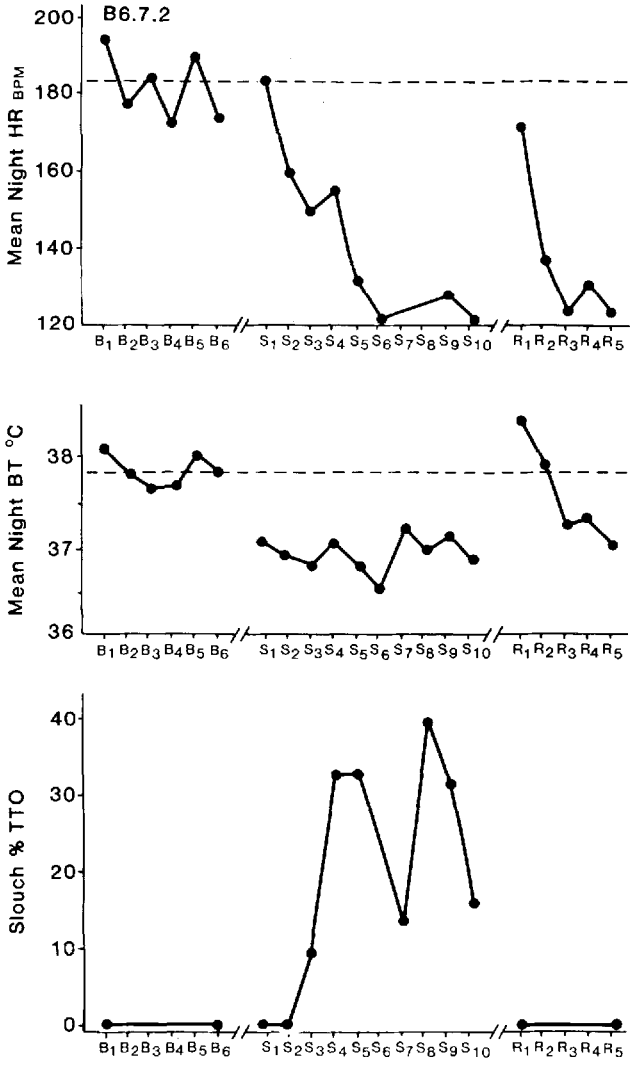


Fig. 2. Mean night heart rate, mean night body temperature, and mean values for daytime slouch behavior for infant B6.7.2 for 6 baseline days, 10 days of separation, and 5 days of reunion.

derive primarily from the degree to which the mother-infant relationship is disrupted by the estrus. If there is a major disruption in the relationship, physiological changes may be seen in the infant. If there is not a major disruption, no systematic physiological changes are seen. In the present case, there is no way we can determine to what extent the mother's being in estrus contributed to the alteration in the mother-infant relationship, compared to, for example, what might have occurred following her return had she not been in estrus. As Hinde and Davies [1972] and Hinde and McGinnis [1977] have pointed out, the mother-infant relationship after reunion is significantly affected by the mother's efforts at reestablishing her social relationships, and this is reflected in the infant's behavior.

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CONCLUSIONS

This study is essentially a case report, and must, therefore, be considered a pilot study. Nonetheless, we believe the study has two major implications. First, by demonstrating that the response to separation in monkeys is not species specific, the general usefulness of the mother-infant separation paradigm in monkeys as an animal model of separation and loss is enhanced. Secondly, the failure of the separated infant to recover physiologically after reunion provided further evidence for the exquisite sensitivity of the infant to the nature of the attachment bond. It may be that the infant's perception of the mother's responsiveness may be a critical factor in this regard. Clearly much more than the mother's physical presence is necessary to completely reverse the separation response. It is also possible, of course, that the relatively long term alterations in heart rate and body temperature reflect separation-induced alterations in autonomic physiological homeostasis that persist beyond the separation per se.

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